

# Technology Stack

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 30 September 2026                                       |
| Team ID       | [ENTER TEAM ID]                                         |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 2 Marks                                                 |

The stack below was chosen for fast development, strong Python AI library support and simple local deployment.

| S.No | Architecture Component / Layer | Technology Chosen                                                                                                                                         | Justification / Purpose                                                                                                                                                           |
|------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | HTML5, CSS3, Jinja2 templates, small vanilla JavaScript                                                                                                   | Server-rendered pages need no build step; embedded CSS gives a responsive, lightweight UI (index, preview, export success).                                                       |
| 2    | Backend / Server-Side          | Python 3, FastAPI, Uvicorn (ASGI), Pydantic                                                                                                               | Fast async framework with automatic /docs, built-in validation, and the same language as the AI libraries.                                                                        |
| 3    | Database / Data Storage        | No database - file storage: static/panels (PNG), static/exports (PDF)                                                                                     | Each comic is generated on demand, so simple files are enough; keeps setup easy. A database is listed as future scope.                                                            |
| 4    | Cloud / Hosting / Deployment   | Local Uvicorn server; Dockerfile provided; GPU via Google Colab / Kaggle if needed                                                                        | Local deployment matches the project brief; Docker makes future cloud hosting straightforward.                                                                                    |
| 5    | Version Control & CI/CD        | Git and GitHub                                                                                                                                            | Required submission platform; supports collaboration and history. Automated tests can be added to GitHub Actions.                                                                 |
| 6    | Third-Party APIs / Other Tools | Google Gemini (google-genai SDK): Flash + Pro; Hugging Face Diffusers, PyTorch, Transformers: Stable Diffusion v1.5; FPDF2; Pillow; python-dotenv; pytest | Gemini Flash creates the structured outline, Gemini Pro writes narration and dialogue, Stable Diffusion draws panels, FPDF2 builds the PDF, Pillow saves images and placeholders. |